



Symposium on **Photonics**  
Emerging Computational Technologies,  
Research and **AI Data Science**

# CALL FOR PAPERS



CONFERENCE DATE

**19-20**

June 2026



ACCEPTANCE NOTIFICATION

**5** June 2026



REGISTRATION DEADLINE

**9** June 2026



<https://ieeespectra.tech>



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## OVERVIEW

SPECTRA 2026 invites submissions at the intersection of photonics and artificial intelligence. Topics of interest include AI-driven photonic device design, optical neural networks, intelligent sensing, photonics for AI hardware acceleration, and machine learning for imaging and spectroscopy. Accepted papers will be presented at the symposium and published in Zenodo.

## SYMPOSIUM SCOPES

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### PHOTONICS & OPTICAL SCIENCE

- Optical Communications & Networks
- Integrated & Silicon Photonics
- Fiber Optics & Photonic Waveguides
- Quantum Photonics & Quantum Information
- Photonic Sensors & Instrumentation
- Imaging & Optical Measurement Systems
- Photonic Materials & Fabrication



### CROSS DISCIPLINARY & FUTURE TECHNOLOGY

- AI-Photonics Convergence
- Computing for Physical Sciences
- Smart Infrastructure & Industry 4.0
- Sustainable Energy & Green Technology
- National Digital Transformation & Smart Governance
- Climate Technology & Disaster Preparedness
- Economic Development Through Technology



### NATURAL SCIENCE ENABLING TECHNOLOGY

- Photonic & Semiconductor Physics
- Material Science for Optoelectronics
- Spectral Analysis & Chemical Sensing
- Computational Physics & Chemistry
- Simulation, Modeling & Optimization
- Experimental Instrumentation & Lab Innovations



### AGRICULTURE, FOOD & ENVIRONMENT

- Smart Agriculture & Precision Farming
- Agricultural Data Science & Remote Sensing
- Climate & Environmental Informatics
- Biosensing & Food Safety Technologies
- Fisheries & Aquaculture Monitoring
- Veterinary Biomedical Technologies



### Business, Economics & Social Systems

- Business Analytics & Data-Driven Decision
- Technology Management & Innovation Strategy
- FinTech, InsurTech & Digital Commerce
- Tech-enabled Supply Chain & Logistics
- ICT for Social Sciences & Public Policy



### EDUCATION, HUMANITIES & COMMUNICATION

- Digital Humanities & Computational Linguistics
- EdTech & Learning Analytics
- Technology Adoption & UX
- Communication in Scientific Dissemination
- Ethics, Policy & Societal Impacts of AI



### HEALTHCARE, MEDICINES & LIFE SCIENCE

- Biomedical Signal & Image Analysis
- Bioinformatics & Computational Biology
- Diagnostic Instrumentation & Wearables
- Assistive & Rehabilitation Technologies



### EMERGING COMPUTATIONAL TECHNOLOGY

- High-Performance Computing (HPC)
- Edge, Fog & Neuromorphic Computing
- Photonic Computing & Optical Processors
- Embedded Systems & VLSI Design
- Internet of Things (IoT) & IIoT
- Microcontroller-Based Applications
- System-on-Chip & Hardware Acceleration



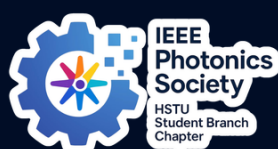
### ARTIFICIAL INTELLIGENCE & DATA SCIENCE

- Machine Learning & Deep Learning
- Data Engineering, Mining & Analytics
- Computer Vision & Perception
- Natural Language & Speech Processing
- AI in Healthcare & Biomedical Apps
- AI in Decision Support Systems
- Human-AI Interaction & Explainable AI

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## SCAN HERE



All selected papers will be included in the **Zenodo Open Repository** and a **DOI** will be provided.